第六章 电磁波的辐射

§6.1 矢势和标势

1. 矢势 / 标势

真空中的 maxwell 方程组：

$$
\begin{cases}
\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} & ① \\
\nabla\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} & ② \\
\nabla\cdot\vec{B} = 0 & ③ \\
\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} & ④
\end{cases}
$$

$\nabla\times(\nabla\times\vec{E}) = \nabla^2\vec{E} - \nabla(\nabla\cdot\vec{E})$

波动方程：

$$\nabla^2\vec{E} - \frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2} = \frac{1}{\varepsilon_0}\nabla\rho + \mu_0\frac{\partial\vec{J}}{\partial t}$$

$$\nabla^2\vec{B} - \frac{1}{c^2}\frac{\partial^2\vec{B}}{\partial t^2} = -\mu_0\nabla\times\vec{J}$$

引入矢势 $\vec{B} = \nabla\times\vec{A}$

④：$\nabla\times\left(\vec{E} + \frac{\partial\vec{A}}{\partial t}\right) = 0$

引入标势 $\vec{E} + \frac{\partial\vec{A}}{\partial t} = -\nabla\varphi$，$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t}$

①：$\nabla^2\varphi + \frac{\partial}{\partial t}(\nabla\cdot\vec{A}) = -\frac{\rho}{\varepsilon_0}$

②：$\nabla^2\vec{A} - \nabla(\nabla\cdot\vec{A}) = \mu_0\vec{J} - \mu_0\varepsilon_0\left(\frac{\partial}{\partial t}\nabla\varphi + \frac{\partial^2\vec{A}}{\partial t^2}\right)$

$$\nabla^2\vec{A} - \frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2} - \nabla\left(\nabla\cdot\vec{A} + \frac{1}{c^2}\frac{\partial\varphi}{\partial t}\right) = -\mu_0\vec{J}$$

例：采用电磁势，表述作用在粒子上的 Lorentz 力

$$\vec{F} = \frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v}\times\vec{B})$$

$$= q\left[-\nabla\varphi - \frac{\partial\vec{A}}{\partial t} + \vec{v}\times(\nabla\times\vec{A})\right]$$

$$\nabla(\vec{v}\cdot\vec{A}) = \vec{v}\times(\nabla\times\vec{A}) + (\vec{v}\cdot\nabla)\vec{A}$$

$$\vec{F} = -q\left[\frac{\partial\vec{A}}{\partial t} + (\vec{v}\cdot\nabla)\vec{A} + \nabla(\varphi - \vec{v}\cdot\vec{A})\right]$$

对流导数 $\frac{d\vec{A}}{dt} = \frac{\partial\vec{A}}{\partial t} + (\vec{v}\cdot\nabla)\vec{A}$

$$d\vec{A}(\vec{r},t) = \frac{\partial\vec{A}}{\partial x}dx + \frac{\partial\vec{A}}{\partial y}dy + \frac{\partial\vec{A}}{\partial z}dz + \frac{\partial\vec{A}}{\partial t}dt$$

$$\frac{d\vec{A}(\vec{r},t)}{dt} = \frac{\partial\vec{A}}{\partial x}v_x + \frac{\partial\vec{A}}{\partial y}v_y + \frac{\partial\vec{A}}{\partial z}v_z + \frac{\partial\vec{A}}{\partial t}$$

$$= (\vec{v}\cdot\nabla)\vec{A} + \frac{\partial\vec{A}}{\partial t}$$

$\rightarrow \frac{d}{dt}(\vec{p} + q\vec{A}) = -\nabla(\varphi - \vec{v}\cdot\vec{A})$

$dP_c = -\nabla W_v$　　P_c：正则动量

∂t

2. 势的规范变换

$$\vec{A} \to \vec{A}' = \vec{A} + \nabla\psi$$

$$\vec{B}' = \nabla\times\vec{A}' = \nabla\times\vec{A} + \nabla\times\nabla\psi = \vec{B}$$

$$\vec{E}' = -\nabla\varphi - \frac{\partial\vec{A}'}{\partial t} = -\nabla\left(\varphi + \frac{\partial\psi}{\partial t}\right) - \frac{\partial\vec{A}}{\partial t}$$

规范变换
$$\begin{cases} \vec{A} \to \vec{A}' = \vec{A} + \nabla\psi \\ \varphi \to \varphi' = \varphi - \frac{\partial\psi}{\partial t} \end{cases}$$

规范变换下，$(\vec{E}, \vec{B})$不变.

规范：选取$(\vec{A}, \varphi)$的"规范"，减少任意性.

3. Lorentz规范和d'Alembert方程.

$$\nabla\cdot\vec{A}_L + \frac{1}{c^2}\frac{\partial\varphi_L}{\partial t} = 0$$

$$\begin{cases} \nabla^2\varphi_L - \frac{1}{c^2}\frac{\partial^2\varphi_L}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \\ \nabla^2\vec{A}_L - \frac{1}{c^2}\frac{\partial^2\vec{A}_L}{\partial t^2} = -\mu_0\vec{J} \end{cases}$$

4. Coulomb规范和势波动方程.

$$\nabla\cdot\vec{A}_C = 0$$

$$\begin{cases} \nabla^2\varphi_C = -\frac{\rho}{\varepsilon_0} \\ \nabla^2\vec{A}_C - \frac{1}{c^2}\frac{\partial^2\vec{A}_C}{\partial t^2} - \frac{1}{c^2}\frac{\partial}{\partial t}\nabla\varphi_C = -\mu_0\vec{J} \end{cases}$$

此时 $\varphi_C = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{x}')}{|\vec{x}-\vec{x}'|}\,dV'$

例：根据d'Alembert方程，求无界自由空间中的行波解.

$$\begin{cases} \nabla^2\varphi_L - \frac{1}{c^2}\frac{\partial^2\varphi_L}{\partial t^2} = 0 \\ \nabla^2\vec{A}_L - \frac{1}{c^2}\frac{\partial^2\vec{A}_L}{\partial t^2} = 0 \end{cases}$$

解得 $\varphi_L = \varphi_{L0}e^{i(\vec{k}\cdot\vec{x} - \omega t)}$ $\quad \vec{A}_L = \vec{A}_{L0}e^{i(\vec{k}\cdot\vec{x} - \omega t)}$

代回Lorentz规范： $i\vec{k}\cdot\vec{A}_{L0}e^{i(\vec{k}\cdot\vec{x}-\omega t)} + \frac{1}{c^2}(-i\omega)\varphi_{L0}e^{i(\vec{k}\cdot\vec{x}-\omega t)} = 0$

$$\vec{k}\cdot\vec{A}_{L0} = \frac{\omega}{c^2}\varphi_{L0}$$

$$\vec{B} = \nabla\times\vec{A} = i\vec{k}\times\vec{A}_L$$

$$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} = -i\vec{k}\varphi_L + i\omega\vec{A}_L$$

$$= -i\vec{k}\frac{c^2}{\omega}\vec{k}\cdot\vec{A}_L + i\omega\vec{A}_L = -i\frac{c^2}{\omega}\left[\vec{k}(\vec{k}\cdot\vec{A}_L) - k^2\vec{A}_L\right]$$

$= -i\frac{c}{\omega}\vec{k}\times(\vec{k}\times\vec{E}) = -\frac{c}{\omega}\vec{k}\times\vec{B}$

§ 6.2 推迟势

1. 含时点电荷的推迟势

在原点处有一点电荷 $Q(t)$：$\rho(\vec{R},t) = Q(t)\delta(\vec{R})$

$$\nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = -\frac{1}{\varepsilon_0}Q(t)\delta(\vec{R})$$

球对称性：$\varphi(\vec{R},t) = \varphi(R,t)$

$$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial\varphi}{\partial R}\right) - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = -\frac{1}{\varepsilon_0}Q(t)\delta(\vec{R})$$

在 $R\neq 0$ 处：$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial\varphi}{\partial R}\right) - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = 0$

令 $\varphi(R,t) = \frac{1}{R}u(R,t)$

$$\frac{\partial^2 u}{\partial R^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0 \quad (R\neq 0)$$

行波法解为 $u(R,t) = f\left(t-\frac{R}{c}\right) + g\left(t+\frac{R}{c}\right)$

$$\varphi(R,t) = \frac{1}{R}f\left(t-\frac{R}{c}\right) + \frac{1}{R}g\left(t+\frac{R}{c}\right)$$ 向内球面波，辐射问题，$=0$

$$= \frac{1}{R}f\left(t-\frac{R}{c}\right)$$

取试探解：$\varphi(R,t) = \frac{1}{4\pi\varepsilon_0}\frac{1}{R}Q\left(t-\frac{R}{c}\right)$

推迟时刻 $t_{ret} = t - \frac{R}{c}$

验证：$\nabla^2\left[\frac{1}{R}Q(t_{ret})\right] = \nabla\cdot\nabla\left\{\frac{1}{R}Q(t_{ret})\right\}$

$= \nabla\cdot\left[\frac{1}{R}\nabla Q(t_{ret}) + Q(t_{ret})\nabla\frac{1}{R}\right]$

$= \frac{1}{R}\nabla^2 Q(t_{ret}) + 2\nabla\frac{1}{R}\cdot\nabla Q(t_{ret}) + Q(t_{ret})\nabla^2\frac{1}{R}$

$= \frac{1}{R}\nabla\cdot\frac{\partial Q}{\partial t_{ret}}\nabla t_{ret} - \frac{2\vec{e}_R}{R^2}\cdot\frac{\partial Q}{\partial t_{ret}}\nabla t_{ret} - 4\pi Q(t_{ret})\delta(\vec{R})$

$\nabla\frac{1}{R} = -\frac{1}{R^2}\vec{e}_R$，

$\nabla^2\frac{1}{R} = -4\pi\delta(\vec{R})$

$\nabla t_{ret} = -\frac{1}{c}\nabla R = -\frac{\vec{e}_R}{c}$

$= -\frac{1}{cR}\nabla\cdot\left(\frac{\partial Q}{\partial t_{ret}}\vec{e}_R\right) + \frac{2}{cR^2}\frac{\partial Q}{\partial t_{ret}} - 4\pi Q(t_{ret})\delta(\vec{R})$

$= -\frac{1}{cR}\left[\nabla\frac{\partial Q}{\partial t_{ret}}\cdot\vec{e}_R + \frac{\partial Q}{\partial t_{ret}}(\nabla\cdot\vec{e}_R)\right] + \frac{2}{cR^2}\frac{\partial Q}{\partial t_{ret}} - 4\pi Q(t_{ret})\delta(\vec{R})$

$= -\frac{1}{cR}\left[\frac{\partial^2 Q}{\partial t_{ret}^2}\nabla t_{ret}\cdot\vec{e}_R + \frac{\partial Q}{\partial t_{ret}}\frac{1}{R^2}\frac{\partial}{\partial R}(R^2)\right] + \frac{2}{cR^2}\frac{\partial Q}{\partial t_{ret}} - 4\pi Q(t_{ret})\delta(\vec{R})$

$= -\frac{1}{cR}\left(\frac{2}{R}\frac{\partial Q}{\partial t_{ret}} - \frac{1}{c}\frac{\partial^2 Q}{\partial t_{ret}^2}\right) + \frac{2}{cR^2}\frac{\partial Q}{\partial t_{ret}} - 4\pi Q(t_{ret})\delta(\vec{R})$

$t_{ret}|_{R=0} = t$

$= \frac{1}{c^2R}\frac{\partial^2 Q}{\partial t_{ret}^2} - 4\pi Q(t)\delta(\vec{R})$

$\nabla^2\left(\frac{Q(t_{ret})}{R}\right) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\frac{Q(t_{ret})}{R} \cdots Q(t)$

$$\nabla \cdot \cdots \frac{1}{4\pi\varepsilon_0 R} \cdots \frac{1}{4\pi\varepsilon_0 R} = -\frac{1}{\varepsilon_0}\delta(R)$$

2. 含时电荷密度分布情况的推迟势

$$\varphi(\vec{R}, t) = \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \frac{\rho(\vec{R}', t_{ret})}{|\vec{R}-\vec{R}'|}$$

其中推迟时间 $t_{ret} = \frac{|\vec{R}-\vec{R}'|}{c}$

例：证明上述标势满足Lorentz规范下的d'Alembert方程。

$$\nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}$$

$$\nabla^2\varphi = \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \, \nabla\cdot\left[(\nabla\tfrac{1}{r})\rho + (\nabla\rho)\tfrac{1}{r}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \left[\rho(\vec{R}', t_{ret})\nabla^2\frac{1}{r} + 2\nabla\rho(\vec{R}', t_{ret})\cdot\nabla\frac{1}{r} + \frac{1}{r}\nabla^2\rho(\vec{R}', t_{ret})\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \left[-4\pi\rho(\vec{R}', t_{ret})\delta(\vec{R}-\vec{R}') + 2\nabla\rho(\vec{R}', t_{ret})\cdot\nabla\frac{1}{r} + \frac{1}{r}\nabla^2\rho(\vec{R}', t_{ret})\right]$$

$$= -\frac{1}{\varepsilon_0}\rho(\vec{R}, t_{ret}) + \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \left[2\nabla\rho(\vec{R}', t_{ret})\cdot\nabla\frac{1}{r} + \frac{1}{r}\nabla^2\rho(\vec{R}', t_{ret})\right]$$

其中 $\int_{V'} dV' \frac{1}{r}\nabla^2\rho(\vec{R}', t_{ret})$ $\quad \frac{\partial\vec{R}'}{\partial\vec{R}} = 0$

$$= \int_{V'} dV' \frac{1}{r}\frac{\partial}{\partial R_i}\left(\frac{\partial\rho}{\partial t_{ret}}\frac{\partial t_{ret}}{\partial R_i}\right)$$ 在直角坐标系中.

$$= \int_{V'} dV' \frac{1}{r}\frac{\partial}{\partial R_i}\left[\frac{\partial\rho}{\partial t_{ret}}\left(-\frac{1}{c}\right)\cdot\frac{1}{r}(R_i - R_i')\right]$$

$$= \frac{1}{c}\int_{V'} dV' \frac{1}{r}\left[\frac{\partial}{\partial R_i}\left(\frac{\partial\rho}{\partial t_{ret}}\right)\frac{R_i'-R_i}{r} + \frac{\partial}{\partial R_i}\left(\frac{1}{r}\right)\frac{\partial\rho}{\partial t_{ret}}(R_i'-R_i) + \frac{\partial(R_i'-R_i)}{\partial R_i}\frac{1}{r}\frac{\partial\rho}{\partial t_{ret}}\right]$$

$$= \frac{1}{c}\int_{V'} dV' \frac{1}{r}\left[\frac{\partial^2\rho}{\partial t_{ret}^2}\frac{\partial t_{ret}}{\partial R_i}\frac{R_i'-R_i}{r} + \frac{1}{r^3}\frac{\partial\rho}{\partial t_{ret}}(R_i'-R_i)^2 - \frac{3}{r}\frac{\partial\rho}{\partial t_{ret}}\right]$$

$$= \frac{1}{c}\int_{V'} dV' \frac{1}{r}\left[\frac{\partial^2\rho}{\partial t_{ret}^2}\frac{(R_i'-R_i)^2}{cr^2} + \frac{1}{r^3}\frac{\partial\rho}{\partial t_{ret}}(R_i'-R_i)^2 - \frac{3}{r}\frac{\partial\rho}{\partial t_{ret}}\right]$$

$$= \frac{1}{c}\int_{V'} dV' \frac{1}{r}\left[\frac{\partial^2\rho}{\partial t_{ret}^2}\frac{(R_i'-R_i)^2}{cr^2} - \frac{2}{r}\frac{\partial\rho}{\partial t_{ret}}\right]$$

$$= \frac{1}{c}\int_{V'} dV' \left(\frac{1}{cr}\frac{\partial^2\rho}{\partial t_{ret}^2} + \frac{2}{r^2}\frac{\partial\rho}{\partial t_{ret}}\right)$$

其中 $\int_{V'} dV' \, 2\nabla\rho(\vec{R}', t_{ret})\cdot\nabla\frac{1}{r}$

$$= 2\int_{V'} dV' \frac{\partial\rho(\vec{R}', t_{ret})}{\partial R_i}\frac{\partial}{\partial R_i}\left(\frac{1}{r}\right) = 2\int_{V'} dV' \frac{\partial\rho}{\partial t_{ret}}\frac{\partial t_{ret}}{\partial R_i}\cdot\frac{R_i'-R_i}{r^3}$$

$$= 2\int_{V'} dV' \frac{\partial\rho}{\partial t_{ret}}\frac{1}{cr^4}(R_i'-R_i)^2 = \frac{2}{c}\int_{V'} dV' \frac{\partial\rho}{\partial t_{ret}}\frac{1}{r^2}$$

代回：

$$\nabla^2\varphi = -\frac{1}{\varepsilon_0}Q(t)\delta(R) + \frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{1}{r}\frac{\partial^2\rho}{\partial t_{ret}^2}$$

而

$$\frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = \cdots$$

$$\frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} = \frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{\partial^2}{\partial t^2}\frac{1}{r}\cdots = \frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{1}{r}\frac{\partial^2 \rho}{\partial t_{ret}^2}$$

类似的 $\vec A(\vec R,t) = \frac{\mu_0}{4\pi}\int_{V'} dV' \frac{1}{r}\vec J(\vec R', t_{ret})$

* 在 Coulomb 规范下，$\varphi_C = \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \frac{\rho(\vec R',t)}{|\vec R-\vec R'|}$

不存在推迟效应.

而矢势满足方程

$$\nabla^2 \vec A_C - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec A_C - \frac{1}{c^2}\frac{\partial}{\partial t}\nabla\varphi_C = -\mu_0\vec J$$

$\vec A_C(\vec R,t)$ 存在推迟效应.

∴ $\varphi_C + \vec A_C$ 存在推迟效应.

例：证明推迟势 $\varphi = \frac{1}{4\pi\varepsilon_0}\int_{V'} dV' \frac{1}{r}\rho(\vec R', t_{ret})$

$$\vec A = \frac{\mu_0}{4\pi}\int_{V'} dV' \frac{1}{r}\vec J(\vec R', t_{ret})$$

满足 Lorentz 规范 $\nabla\cdot\vec A(\vec R,t) + \frac{1}{c^2}\frac{\partial}{\partial t}\varphi(\vec R,t) = 0$

$$\nabla\cdot\vec A = \frac{\mu_0}{4\pi}\int_{V'} dV'\, \nabla\cdot\left[\frac{1}{r}\vec J(\vec R', t_{ret})\right]$$

$$= \frac{\mu_0}{4\pi}\int_{V'} dV' \left[\nabla\frac{1}{r}\cdot\vec J(\vec R', t_{ret}) + \frac{1}{r}\nabla\cdot\vec J(\vec R', t_{ret})\right]$$

$$= \frac{\mu_0}{4\pi}\int_{V'} dV' \left[-\nabla'\frac{1}{r}\cdot\vec J(\vec R', t_{ret}) + \frac{1}{r}\frac{\partial\vec J(\vec R', t_{ret})}{\partial t_{ret}}\cdot\nabla t_{ret}\right] \qquad \nabla f(r) = -\nabla' f(r)$$

$$= \frac{\mu_0}{4\pi}\int_{V'} dV' \left[-\nabla'\frac{1}{r}\,\vec J(\vec R', t_{ret}) - \frac{1}{r}\frac{\partial\vec J(\vec R', t_{ret})}{\partial t_{ret}}\cdot\nabla' t_{ret}\right] \qquad t_{ret} = t - \frac{r}{c}$$

$$\frac{1}{c^2}\frac{\partial\varphi}{\partial t} = \frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{1}{r}\frac{\partial\rho(\vec R', t_{ret})}{\partial t}$$

$$= \frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{1}{r}\frac{\partial\rho(\vec R', t_{ret})}{\partial t_{ret}} \qquad \nabla\cdot\vec J + \frac{\partial\rho}{\partial t} = 0$$

$$= -\frac{1}{4\pi\varepsilon_0 c^2}\int_{V'} dV' \frac{1}{r}\nabla'\cdot\vec J(\vec R', t_{ret})\Big|_{t_{ret}} \qquad \nabla'\cdot\vec J(\vec R', t_{ret})\Big|_{t_{ret}} + \frac{\partial\rho(\vec R', t_{ret})}{\partial t_{ret}} = 0$$

$$\nabla\cdot\vec A + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} = \frac{\mu_0}{4\pi}\int_{V'}\Big[-\nabla'\frac{1}{r}\,\vec J(\vec R', t_{ret}) - \frac{1}{r}\frac{\partial\vec J(\vec R', t_{ret})}{\partial t_{ret}}\cdot\nabla' t_{ret} - \frac{1}{r}\nabla'\cdot\vec J(\vec R', t_{ret})\Big|_{t_{ret}}\Big]$$

$$= \frac{\mu_0}{4\pi}\int_{V'}\left[-\nabla'\frac{1}{r}\cdot\vec J(\vec R', t_{ret}) - \frac{1}{r}\nabla'\cdot\vec J(\vec R', t_{ret})\right]$$

$$= -\frac{\mu_0}{4\pi}\int_{V'} dV'\,\nabla'\cdot\left[\frac{1}{r}\vec J(\vec R', t_{ret})\right]$$

$$= -\frac{\mu_0}{4\pi}\oint d\vec S'\cdot\frac{1}{r}\vec J(\vec R', t_{ret}) = 0$$ (封闭曲面)

3. 时谐电流的无限长的直导线的势和电磁场

$$\vec{A}(\vec{R},t) = \frac{\mu_0}{4\pi}\int_{V'} dV' \frac{1}{r}\vec{J}(\vec{R}',t_{ret}) = \frac{\mu_0}{4\pi}\int_{-\infty}^{\infty} dz \frac{1}{r} I(z,t_{ret})\,\vec{e}_z$$

$$= \frac{\mu_0 I_0}{4\pi}\int_{-\infty}^{\infty} dz \frac{1}{r} e^{-i\omega t} e^{i\omega \frac{r}{c}}\,\vec{e}_z \qquad I(t) = I_0 e^{-i\omega t}$$

$\vec{A}(\vec{R},t) = A_z\,\vec{e}_z$ 旋转对称性、

$$A_z(R,t) = \frac{\mu_0 I_0}{4\pi}\int_{-\infty}^{\infty} dz \frac{1}{\sqrt{z^2+R^2}} e^{-i\omega t} e^{i\omega\frac{(z^2+R^2)^{1/2}}{c}}$$

$$= \frac{\mu_0 I_0}{4\pi} e^{-i\omega t}\int_{-\infty}^{\infty} \frac{1}{\sqrt{z^2+R^2}} e^{ik_0(z^2+R^2)^{1/2}} \qquad \frac{\omega}{c} = k_0 = \frac{2\pi}{\lambda_0}$$

$R \ll \lambda_0$ 时、在 $|z| \ll R$ 区域，$e^{ik_0(z^2+R^2)^{1/2}} \approx 1$，无推迟现象.

$R \gg \lambda_0$ 时，在 $|z| \ll R$ 区域

$$A_z(R,t) \approx \frac{\mu_0 I_0}{4\pi} e^{-i\omega t}\int_{-\infty}^{\infty} dz \frac{1}{R}\left(1-\frac{z^2}{2R^2}\right) e^{ik_0 R\left(1+\frac{z^2}{2R^2}\right)}$$

$$\approx \frac{\mu_0 I_0}{4\pi R} e^{i(k_0 R-\omega t)} \underline{\int_{-\infty}^{\infty} dz\, e^{i\frac{1}{2}k_0\frac{z^2}{R}}} \qquad \text{在 } |z| \ll R \text{ 时为主要贡献}$$

Gauss 积分 $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$ $\quad \int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$

$$= \frac{\mu_0 I_0}{4\pi R} e^{i(k_0 R-\omega t)}\sqrt{\frac{\pi}{\frac{k_0}{2Ri}}} = \frac{\mu_0 I_0}{\sqrt{8\pi k_0 R}} e^{i(k_0 R-\omega t+\frac{\pi}{4})}$$

$\vec{B}(\vec{R},t) = \nabla\times\vec{A}(\vec{R},t)$ 柱坐标 $\nabla = \left\langle \frac{\partial}{\partial R}, \frac{1}{R}\frac{\partial}{\partial\varphi}, \frac{\partial}{\partial z}\right\rangle$

$$= -\frac{\mu_0 I_0}{\sqrt{8\pi k_0}}\frac{\partial}{\partial R}\left[\frac{1}{\sqrt{R}} e^{i(k_0 R-\omega t+\frac{\pi}{4})}\right]\vec{e}_\varphi$$

$$= \frac{\mu_0 I_0}{\sqrt{8\pi k_0}}\frac{1}{R}\left(\frac{1}{2\sqrt{R}} - ik_0\sqrt{R}\right) e^{i(k_0 R-\omega t+\frac{\pi}{4})}\,\vec{e}_\varphi$$

$$\approx \mu_0 I_0\sqrt{\frac{k_0}{8\pi R}} e^{i(k_0 R-\omega t-\frac{\pi}{4})}\,\vec{e}_\varphi \qquad R \gg \lambda_0 = \frac{2\pi}{k_0}$$

$\nabla\times\vec{B} = \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}$

$$\frac{\partial\vec{E}}{\partial t} = -i\omega\vec{E} = \frac{1}{\mu_0\varepsilon_0}\mu_0 I_0\sqrt{\frac{k_0}{8\pi}}\frac{\partial}{\partial R}\left[\frac{1}{\sqrt{R}} e^{i(k_0 R-\omega t-\frac{\pi}{4})}\right]\vec{e}_z$$

$$= \frac{I_0}{\varepsilon_0}\sqrt{\frac{k_0}{8\pi}}\frac{1}{R}\left(ik_0\sqrt{R} - \frac{1}{2}\frac{1}{\sqrt{R}}\right) e^{i(k_0 R-\omega t-\frac{\pi}{4})}\,\vec{e}_z \qquad \frac{k_0}{\omega} = \frac{1}{c} = \sqrt{\varepsilon_0\mu_0}$$

$$\vec{E} \approx -\sqrt{\frac{\mu_0}{\varepsilon_0}} I_0\sqrt{\frac{k_0}{8\pi R}} e^{i(k_0 R-\omega t-\frac{\pi}{4})}\,\vec{e}_z$$

$$\vec{S} = \vec{E}\times\vec{H} = Re(\vec{E})\times\frac{1}{\mu_0}Re(\vec{B}) = \sqrt{\frac{\mu_0}{\varepsilon}}\frac{I_0^2 k_0}{\ldots}\cos^2(k_0 R-\omega t-\frac{\pi}{4})\,\vec{e}_R$$

$$\langle \vec S\rangle = \sqrt{\frac{\mu_0}{\varepsilon_0}}\,\frac{I_0^2 k_0}{16\pi R}\,\vec e_R$$

$$P = 2\pi R\,|\langle \vec S\rangle| = \frac{1}{8}\sqrt{\frac{\varepsilon_0}{\mu_0}}\,I_0^2 k_0 \frac{1}{R}$$

4. 无限大平面上时谐面电流的电磁场.

设平面上 $\vec K(t) = \vec K_0 e^{-i\omega t} = K_0 e^{-i\omega t}\vec e_x$

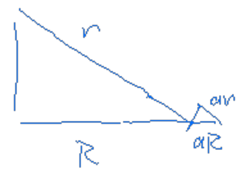
在 $z \gg \lambda_0$ 区域,

$$\vec A(\vec R, t) = \frac{\mu_0}{4\pi}\int_{S'} dS' \frac{1}{r}\vec K(\vec R', t_{ret})$$

$$A_x(z,t) = \frac{\mu_0}{4\pi}\int_{S'} dx\,dy\,\frac{1}{r} K_0 e^{-i\omega t} e^{i\omega\frac{r}{c}}$$

$$= \frac{\mu_0 K_0}{4\pi} e^{-i\omega t}\int_{S'} R\,dR\,d\phi\,\frac{1}{r} e^{i\omega\frac{r}{c}} \qquad \frac{dR}{r} = \frac{dr}{R}$$

$$= \frac{\mu_0 K_0}{2} e^{-i\omega t}\int_{|z|}^{\infty} e^{i\omega\frac{r}{c}}\,dr$$



为保证积分收敛:

$$A_x(z,t) = \lim_{\lambda\to 0}\frac{\mu_0 K_0}{2} e^{-i\omega t}\int_{|z|}^{\infty} e^{-\lambda r} e^{i\omega\frac{r}{c}}\,dr$$

$$= \lim_{\lambda\to 0} -\frac{\mu_0 K_0}{2} e^{-i\omega t}\frac{1}{i\frac{\omega}{c}-\lambda} e^{(i\frac{\omega}{c}-\lambda)|z|}$$

$$= \frac{\mu_0 K_0}{2k_0} e^{i(k_0|z| - \omega t + \frac{\pi}{2})}$$

$$\vec B(z,t) = \nabla\times\vec A(z,t) = \frac{\partial A_x}{\partial z}\vec e_y$$

$$= \frac{\mu_0 K_0}{2k_0} e^{i(k_0|z| - \omega t + \pi)}\,\mathrm{sgn}(z)\,\vec e_y$$

$$= \begin{cases} -\dfrac{\mu_0 K_0}{2} e^{i(k_0 z - \omega t)}\vec e_y & (z>0) \\ \dfrac{\mu_0 K_0}{2} e^{i(-k_0 z - \omega t)}\vec e_y & (z<0)\end{cases}$$

$$\frac{k_0}{\omega} = \sqrt{\varepsilon_0\mu_0}$$

$$\vec E(z,t) = \frac{\nabla\times\vec B(z,t)}{-i\omega\mu_0\varepsilon_0} \qquad -\frac{1}{i\omega\mu_0\varepsilon_0}(ik_0)\frac{1}{2}\mu_0 K_0 = -\frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$= \begin{cases} -\frac{1}{2}K_0 Z_0\, e^{i(k_0 z - \omega t)}\vec e_x & (z>0) \\ -\frac{1}{2}K_0 Z_0\, e^{i(-k_0 z - \omega t)}\vec e_x & (z<0)\end{cases}$$

例:用无限大平面电流的电磁场解释为何电磁波入射时,理想导体内部无电磁场

入射波 $\vec E_I = \vec E_{I0} e^{i(-k_0 z - \omega t)}\vec e_x$

反射波 $\vec E_R' = \vec E_{R0}' e^{i(k_0 z - \omega t)}$ $\quad \vec E_{I0} + \vec E_{R0}' = 0$

表面磁场: $\vec H = -(H_{I0} + H_{R0}')e^{-i\omega t}\vec e_y$

$$= -\frac{2}{Z_0}E_{I0}e^{-i\omega t}\vec e_y$$

$\vec{B} = \vec{n}\times\vec{H} = \vec{e_z}\times\vec{H} = \frac{\varepsilon}{\varepsilon_0}\vec{E_{z0}}e^{-i\omega t}\vec{e_x}$

相应辐射电场，$\vec{E_{rad}} = \begin{cases} -\vec{E_{z0}}e^{i(k_0 z-\omega t)} = \vec{E_R} & z>0 \\ -\vec{E_{z0}}e^{i(-k_0 z-\omega t)} \text{ 与入射波抵消，} & z<0 \end{cases}$

§ 6.3 时谐电流的多极辐射

1. 时谐电流场的推迟势

$\vec{J}(\vec{R}',t) = \vec{J}(\vec{R}')e^{-i\omega t}$

$\vec{A}(\vec{R}',t) = \frac{\mu_0}{4\pi}\int_{V'}\frac{1}{r}\vec{J}(\vec{R}',t-\frac{r}{c}) = \frac{\mu_0}{4\pi}\int_{V'}dV'\,\vec{J}(\vec{R}')e^{-i\omega(t-\frac{r}{c})}$

$= \frac{\mu_0}{4\pi}\int_{V'}\frac{1}{r}\vec{J}(\vec{R}')\underbrace{e^{ik_0 r}}_{\text{推迟作用因子}}e^{-i\omega t} = \vec{A}(\vec{R})e^{-i\omega t}$

$\varphi(\vec{R},t) = \varphi(\vec{R})e^{-i\omega t}$

Lorentz规范，$\nabla\cdot\vec{A}(\vec{R}) = i\frac{\omega}{c^2}\varphi(\vec{R})$

2. 推迟势的多极展开

（电流分布区域线度 $l \ll r \ll$ 辐射波长 λ_0）

在远场条件下 $l \ll \lambda_0 \ll r$

$r = |\vec{R}-\vec{R}'| = (R^2 - 2\vec{R}\cdot\vec{R}' + R'^2)^{1/2}$

$\approx R\left(1-\frac{2\vec{R}'\cdot\vec{e_R}}{R}\right)^{1/2} \approx R - \vec{R}'\cdot\vec{e_R}$

$\vec{A}(\vec{R}) = \frac{\mu_0}{4\pi}\int_{V'}dV'\frac{\vec{J}(\vec{R}')e^{ik_0(R-\vec{e_R}\cdot\vec{R}')}}{R-\vec{e_R}\cdot\vec{R}'}$

$\approx \frac{\mu_0}{4\pi}\int_{V'}dV'\,\vec{J}(\vec{R}')\frac{e^{ik_0 R}}{R}(1-e^{ik_0\vec{e_R}\cdot\vec{R}'})(1+\frac{\vec{e_R}\cdot\vec{R}'}{R})$

$\approx \frac{\mu_0}{4\pi}\int_{V'}dV'\,\vec{J}(\vec{R}')\frac{e^{ik_0 R}}{R}(1-ik_0\vec{e_R}\cdot\vec{R}')$ $\quad e^{ik_0(\vec{e_R}\cdot\vec{R}')} \approx \frac{l}{\lambda_0}$，$\frac{\vec{e_R}\cdot\vec{R}'}{R} \approx \frac{l}{r}$

$= \frac{\mu_0}{4\pi}\frac{e^{ik_0 R}}{R}\int_{V'}dV'\,\vec{J}(\vec{R}') - \frac{\mu_0}{4\pi}\frac{e^{ik_0 R}}{R}ik_0\int_{V'}dV'\,\vec{J}(\vec{R}')\,\vec{e_R}\cdot\vec{R}'$

$= \vec{A}^{(0)}(\vec{R}) + \vec{A}^{(1)}(\vec{R})$

3. 时谐电偶极子的推迟势

张量的 Gauss 定理：$\vec{n}\cdot\vec{J}\Big|_{S_{V'}} = 0$

$$\int_{V'} dV'\,\vec{J} = -\int_{V'} dV'(\nabla'\cdot\vec{J})\vec{R}'$$

$$\rho(\vec{R}',t) = \rho(\vec{R}')e^{-i\omega t}$$

连续性方程：$\nabla'\cdot\vec{J}(\vec{R}',t) + \frac{\partial\rho(\vec{R}',t)}{\partial t} = 0 \rightarrow \nabla'\cdot\vec{J}(\vec{R}') = i\omega\rho(\vec{R}')$

$$\int_{V'} dV'\,\vec{J} = -\int_{V'} dV'\, i\omega\rho(\vec{R}')\vec{R}' = -i\omega\underline{\vec{p}}$$

电偶极矩振幅

$$\vec{p}(\vec{R},t) = \vec{p}(\vec{R})e^{-i\omega t}$$

$$= \frac{i}{\omega}\int_{V'} dV'\,\vec{J}(\vec{R}') = \frac{i}{\omega}\int_{L} I_0\, d\vec{l}'$$

$$\Rightarrow \vec{A}^{(0)}(\vec{R}) = \frac{\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\dot{\vec{p}}(\vec{R},t) \qquad \dot{\vec{p}}(\vec{R},t) = -i\omega\vec{p} = \int_{V} dV'\,\vec{J}(\vec{R}')$$

4. 时谐电偶极子的辐射场及辐射功率

$$\begin{cases}\vec{B}^{(0)}(\vec{R},t) = \nabla\times\vec{A}^{(0)}(\vec{R},t)\\ \vec{E}^{(0)}(\vec{R},t) = \frac{ic}{k_0}\nabla\times\vec{B}(\vec{R},t)\end{cases}$$

$$\nabla\left(\frac{e^{ik_0R}}{R}\right) = ik_0\frac{e^{ik_0R}}{R}\vec{e}_R + e^{ik_0R}\nabla\left(\frac{1}{R}\right)$$

$$= \left(ik_0 - \frac{1}{R}\right)\frac{e^{ik_0R}}{R}\vec{e}_R \qquad (r \gg \lambda_0)$$

$$\approx ik_0\frac{e^{ik_0R}}{R}\vec{e}_R$$

$$\vec{B}^{(0)}(\vec{R},t) = ik_0\vec{e}_R\times\vec{A}^{(0)}(\vec{R},t)$$

$$= \frac{\mu_0}{4\pi}\frac{e^{ik_0R}}{R}(ik_0\vec{e}_R)\times\dot{\vec{p}}(\vec{R},t) \qquad \dot{\vec{p}} = -i\omega\vec{p}$$

$$= \frac{\mu_0}{4\pi c}\frac{e^{ikR}}{R}\ddot{\vec{p}}(\vec{R},t)\times\vec{e}_R$$

电荷加速运动才会辐射。

设 $\vec{p} = p\vec{e}_z$

$$\vec{B}^{(0)}(\vec{R},t) = -\frac{\mu_0\omega^2}{4\pi c}\frac{e^{i(k_0R-\omega t)}}{R}p\sin\theta\,\vec{e}_\phi \qquad \vec{e}_z\times\vec{e}_R = \sin\theta\,\vec{e}_\phi$$

$$\vec{E}^{(0)}(\vec{R},t) = -c\vec{e}_R\times\vec{B}^{(0)}(\vec{R},t) = -\frac{\mu_0\omega^2}{4\pi}\frac{e^{i(k_0R-\omega t)}}{R}p\sin\theta\,\vec{e}_\theta$$

$$\vec{S} = \mathrm{Re}(\vec{E}^{(0)})\times\frac{1}{\mu_0}\mathrm{Re}(\vec{B}^{(0)})$$

$$= \frac{\mu_0}{16\pi^2 c} \frac{}{R^2} \sin\theta \cos(\ldots - \omega \ldots) \vec{e}_R$$

$$\langle \vec{S} \rangle = \frac{\mu_0}{32\pi^2 c} \frac{\omega^4 p^2}{R^2} \sin^2\theta \, \vec{e}_R$$

$$P = \oint |\langle \vec{S} \rangle| \, d\Omega = \frac{\mu_0}{32\pi^2 c} \frac{\omega^4 p^2}{R^2} \oint \sin^2\theta \, d\Omega = \frac{\mu_0}{12\pi c} \omega^4 p^2$$

例：一中心馈电偶直线天线，电流分布满足：

$$I_z(t) = I_0 \left(1 - \frac{2}{L}|z|\right) e^{-i\omega t}$$

且 $\omega L \ll c$，采用电偶极子辐射近似，计算其在远场的辐射功率。

$$\vec{p} = \frac{i}{\omega} \int_{-\frac{L}{2}}^{\frac{L}{2}} I_z(t) \, dz$$

$$= \frac{i I_0 \cos e^{-i\omega t}}{\omega} \int_{-\frac{L}{2}}^{\frac{L}{2}} \left(1 - \frac{2|z|}{L}\right) dz$$

$$= \frac{i I_0 e^{-i\omega t}}{\omega} \left(L - \frac{1}{L} \cdot \frac{1}{2} L^2\right) = i \frac{I_0 L e^{-i\omega t}}{2\omega}$$

$$P = \frac{\mu_0}{12\pi c} |\vec{p}|^2 \omega^4 = \frac{\mu_0 I_0^2 L^2}{48\pi c} \omega^2$$

定义辐射电阻：$P = \frac{1}{2} I_0^2 R$

解得 $R = \frac{\mu_0 L^2}{24\pi c} \omega^2 = \frac{\pi}{6} Z_0 \left(\frac{L}{\lambda_0}\right)^2$ $\qquad \omega = c k_0 = \frac{2\pi c}{\lambda_0}$

G. 时谐磁偶极子的辐射场

$$\vec{A}^{(1)}(\vec{R}) = -\frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \int_{V'} dV' (\vec{e}_R \cdot \vec{R}') \vec{J}(\vec{R}')$$

$$= -\frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \vec{e}_R \cdot \int_{V'} dV' \vec{R}' \vec{J}(\vec{R}')$$

$$= -\frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \vec{e}_R \cdot \int_{V'} \left[\frac{1}{2}(\vec{R}'\vec{J} - \vec{J}\vec{R}') + \frac{1}{2}(\vec{R}'\vec{J} + \vec{J}\vec{R}')\right]$$

磁偶极子辐射 $= \underline{\vec{A}_m^{(1)}(\vec{R})} + \underline{\vec{A}_Q(\vec{R})}$ 电四极子辐射。

$$\vec{A}_m^{(1)}(\vec{R}) = -\frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \vec{e}_R \cdot \int_{V'} dV' \frac{1}{2} (\vec{R}'\vec{J} - \vec{J}\vec{R}')$$

$$= -\frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \frac{1}{2} \int_{V'} dV' \left[(\vec{e}_R \cdot \vec{R}')\vec{J} - (\vec{e}_R \cdot \vec{J})\vec{R}'\right]$$

$$= \frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \vec{e}_R \times \frac{1}{2} \int_{V'} (\vec{R}' \times \vec{J}) \, dV'$$

$$= \frac{i k_0 \mu_0}{4\pi} \frac{e^{ik_0 R}}{R} \vec{e}_R \times \vec{m}$$ 磁矩 $\vec{m} = \frac{1}{2} \int_{V'} dV' (\vec{R}' \times \vec{J})$

$$\vec{A}^{(1)}(\vec{R}, t) = \vec{A}^{(1)}(\vec{R}) e^{-i\omega t}$$

$$\vec{A}_m^{(1)} = \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\vec{e}_R\times\vec{m}\,e^{-i\omega t}$$

$$= \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\vec{e}_R\times\vec{m}(t)$$

$$\vec{B}_m^{(1)}(\vec{R},t) = \nabla\times\vec{A}_m^{(1)}(\vec{R},t)$$

$$= -\frac{k_0^2\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\vec{e}_R\times\left[\vec{e}_R\times\vec{m}(t)\right]$$

$$= \frac{\mu_0}{4\pi c^2}\frac{e^{ik_0R}}{R}\vec{e}_R\times\left[\vec{e}_R\times\ddot{\vec{m}}(t)\right] \qquad \ddot{\vec{m}}(t) = -\omega^2\vec{m}(t) = -k_0^2c^2\vec{m}(t)$$

$$\vec{E}_m^{(1)}(\vec{R},t) = -c\,\vec{e}_R\times\vec{B}_m^{(1)}(\vec{R},t)$$

$$= -\frac{\mu_0}{4\pi c}\frac{e^{ik_0R}}{R}\vec{e}_R\times\ddot{\vec{m}}(t) \qquad \vec{e}_R\times\vec{e}_R\times\vec{B} = -\vec{B}$$

设 $\vec{m} = m\vec{e}_z$

$$\vec{B}_m^{(1)}(\vec{R},t) = \frac{\mu_0}{4\pi c^2}\frac{e^{ik_0R}}{R}\ddot{m}(t)\sin^2\theta\,\vec{e}_\theta$$

$$\vec{E}_m^{(1)}(\vec{R},t) = \frac{\mu_0}{4\pi c}\frac{e^{ik_0R}}{R}\ddot{m}(t)\sin\theta\,\vec{e}_\phi$$

$$\vec{S} = \mathrm{Re}\left[\vec{E}_m^{(1)}(\vec{R},t)\right]\times\frac{1}{\mu_0}\mathrm{Re}\left[\vec{B}_m^{(1)}(\vec{R},t)\right]$$

$$= \frac{\mu_0}{16\pi^2c^3}\frac{m^2\omega^4}{R^2}\sin^2\theta\cos^2(k_0R-\omega t)\,\vec{e}_R$$

$$P_m = \oint\langle\vec{S}\rangle\,d\Omega = \frac{\mu_0}{12\pi c^3}m^2\omega^4$$

例：有一载流圆线圈，线圈中电流为 $I(t) = I_0\cos(\omega t)$，线圈半径为 a，满足 $\omega a \ll c$，采用磁偶极子近似，计算其在远场的辐射功率。

$$\vec{p} = 0 \rightarrow \vec{A}^{(0)} = 0$$

$$\vec{m} = I\hat{S} = I_0\pi a^2\cos(\omega t)\,\hat{S}$$

$$P = \frac{\mu_0}{12\pi c^3}I_0^2\pi^2a^4\omega^4 = \pi\frac{\mu_0}{12c^3}I_0^2a^4\omega^4$$

$$R = \frac{2P}{I_0^2} = \frac{\pi\mu_0}{6c^3}a^4\omega^4 \qquad \omega = \frac{2\pi c}{\lambda}$$

$$= \frac{8}{3}\pi^3 Z_0\left(\frac{a}{\lambda_0}\right)^4$$

$R_m^{(1)} \ll R^{(0)} \rightarrow$ 磁偶极子辐射 $\ll$ 电偶极子辐射。

6. 时谐电四极子的辐射场

$$\int_{V'} dV' (\vec{J}\vec{R}' + \vec{R}'\vec{J}) = -i\omega \int_{V'} \rho \vec{R}'\vec{R}'$$

$$\nabla' \cdot \vec{J}(\vec{R}') = i\omega\rho(\vec{R}')$$

$$\int_V dV (\vec{J}\vec{r} + \vec{r}\vec{J}) = -\int_V dV (\nabla \cdot \vec{J})\vec{r}\vec{r} \quad (\vec{n}\cdot\vec{J}|_{\partial V} = 0)$$

$$\vec{A}_D^{(1)}(\vec{R}) = -\frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\vec{e}_R \cdot \int_{V'} dV' \frac{1}{2}(\vec{R}'\vec{J} + \vec{J}\vec{R}')$$

$$= \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}(i\omega)\,\vec{e}_R \cdot \int_{V'} dV' \frac{1}{2}\rho(\vec{R}')\vec{R}'\vec{R}'$$

$$= \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}(i\omega\,\vec{e}_R \cdot \overleftrightarrow{D})$$

$$\vec{A}_D^{(1)}(\vec{R},t) = \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}(i\omega\,\vec{e}_R \cdot \overleftrightarrow{D} e^{-i\omega t})$$

$$= \frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\, i\omega\,\vec{e}_R \cdot \overleftrightarrow{D}(t)$$

$$\dddot{\overleftrightarrow{D}}(t) = (-i\omega)^3\overleftrightarrow{D}(t) = i\omega^3\overleftrightarrow{D}(t)$$

$$= -\frac{ik_0\mu_0}{4\pi}\frac{e^{ik_0R}}{R}\vec{e}_R \cdot \dot{\overleftrightarrow{D}}(t)$$

$$= \frac{\mu_0}{4\pi c}\frac{e^{ik_0R}}{R}\vec{e}_R \cdot \ddot{\overleftrightarrow{D}}(t)$$

$$\vec{B}_D^{(1)}(\vec{R},t) = \nabla \times \vec{A}_D^{(1)}(\vec{R},t) = -\frac{\mu_0}{4\pi c^2}\frac{e^{ik_0R}}{R}\vec{e}_R \times [\vec{e}_R \cdot \dddot{\overleftrightarrow{D}}(t)]$$

$$\vec{E}_D^{(1)}(\vec{R},t) = \frac{1}{-i\omega\mu_0\varepsilon_0}\nabla \times \vec{B}_D^{(1)}(\vec{R},t) = \frac{\mu_0}{4\pi c^2}\frac{e^{ik_0R}}{R}\vec{e}_R \times \{\vec{e}_R \times [\vec{e}_R \times \dddot{\overleftrightarrow{D}}(t)]\}$$

$$= -\frac{\mu_0}{4\pi c}\frac{e^{ik_0R}}{R}\vec{e}_R \cdot \ddot{\overleftrightarrow{D}}(t)$$

设 $\overleftrightarrow{D}(t) = \overleftrightarrow{D}\,\vec{e}_z$

$$\vec{S} = \mathrm{Re}[\vec{E}_D^{(1)}(\vec{R},t)] \times \frac{1}{\mu_0}\mathrm{Re}[\vec{B}_D^{(1)}(\vec{R},t)]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{8\pi c^5}\frac{1}{R^2}\left[[\vec{e}_R \cdot \dddot{\overleftrightarrow{D}}(t)] \times \vec{e}_R\right]^2 \vec{e}_R$$

例：位于 xoy 平面中心的线段，长为 a，以恒定角速度 ω 绕 z 轴旋转，

线段两端各有一个电荷为 q 的点电荷，若 $\omega a \ll c$，

(1) 计算体系的电偶极矩，磁偶极矩，电四极矩。

$$\vec{p} = q\vec{r} - q\vec{r} = 0$$

$$\vec{m} = I\vec{S} = \frac{2q}{\frac{2\pi}{\omega}} \pi \left(\frac{a}{2}\right)^2 \vec{e_z} = \frac{1}{4} q\omega a^2 \vec{e_z}$$

$$\overleftrightarrow{D} = \frac{1}{2} \cdot 2q \left(\frac{a}{2}\right)^2 \left[\cos(\omega t)\vec{e_x} + \sin(\omega t)\vec{e_y}\right]^2$$

$$= \frac{1}{4} q a^2 \left[\cos^2(\omega t)\vec{e_x}\vec{e_x} + \frac{1}{2}\sin(2\omega t)(\vec{e_x}\vec{e_y} + \vec{e_y}\vec{e_x}) + \cos^2(\omega t)\vec{e_y}\vec{e_y}\right]$$

$$= \frac{1}{8} q a^2 \left\{\left[1+\cos(2\omega t)\right]\vec{e_x}\vec{e_x} + \sin(2\omega t)(\vec{e_x}\vec{e_y} + \vec{e_y}\vec{e_x}) + \left[1+\cos(2\omega t)\right]\vec{e_y}\vec{e_y}\right\}$$

(2) 说明电场辐射的类型，给出辐射频率。

电四极，2ω

(3) 在远场下 (R, θ, φ) 处观测，分析沿着轴 $\theta = 0^\circ$，$\theta = 90^\circ$ 的辐射场的偏振状态

$$\begin{cases} \vec{e_R} = \sin\theta\cos\varphi\,\vec{e_x} + \sin\theta\sin\varphi\,\vec{e_y} + \cos\theta\,\vec{e_z} \\ \vec{e_\theta} = \cos\theta\cos\varphi\,\vec{e_x} + \cos\theta\sin\varphi\,\vec{e_y} - \sin\theta\,\vec{e_z} \\ \vec{e_\varphi} = -\sin\varphi\,\vec{e_x} + \cos\varphi\,\vec{e_y} \end{cases}$$

$$\vec{e_R} \cdot \dddot{\overleftrightarrow{D}}(t)$$

$$= (\sin\theta\cos\varphi\,\vec{e_x} + \sin\theta\sin\varphi\,\vec{e_y} + \cos\theta\,\vec{e_z}) \cdot \frac{d^3}{dt^3}$$

$$\frac{1}{8} q a^2 \left\{\left[1+\cos(2\omega t)\right]\vec{e_x}\vec{e_x} + \sin(2\omega t)(\vec{e_x}\vec{e_y} + \vec{e_y}\vec{e_x}) + \left[1-\cos(2\omega t)\right]\vec{e_y}\vec{e_y}\right\}$$

$$= q\omega^3 a^2 (\sin\theta\cos\varphi\,\vec{e_x} + \sin\theta\sin\varphi\,\vec{e_y} + \cos\theta\,\vec{e_z})$$

$$\left[\sin(2\omega t)\vec{e_x}\vec{e_x} - \cos(2\omega t)(\vec{e_x}\vec{e_y} + \vec{e_y}\vec{e_x}) - \sin(2\omega t)\vec{e_y}\vec{e_y}\right]$$

$$= q\omega^3 a^2 \left\{\sin\theta\cos\varphi\left[\sin(2\omega t)\vec{e_x} - \cos(2\omega t)\vec{e_y}\right] - \sin\theta\sin\varphi\left[\cos(2\omega t)\vec{e_x} + \sin(2\omega t)\vec{e_y}\right]\right\}$$

$$= q\omega a \quad \theta \quad \sin(2\omega t-\varphi)\,\vec{e}_x - \cos(2\omega t-\varphi)\,\vec{e}_y$$

$$\left[\vec{e}_R\cdot\dddot{\vec{D}}(t)\right]\times\vec{e}_R = q\omega^3 a^2\sin\theta\left[\sin(2\omega t-\varphi)\,\vec{e}_x - \cos(2\omega t-\varphi)\,\vec{e}_y\right]\times$$

$$(\sin\theta\cos\varphi\,\vec{e}_x + \sin\theta\sin\varphi\,\vec{e}_y + \cos\theta\,\vec{e}_z)$$

$$= q\omega^3 a^2\sin\theta\left[-\cos\theta\cos(2\omega t-\varphi)\,\vec{e}_x - \cos\theta\sin(2\omega t-\varphi)\,\vec{e}_y + \sin\theta\cos(2\omega t-2\varphi)\,\vec{e}_z\right]$$

$$A = \begin{pmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \cos\theta\cos\varphi & \cos\theta\sin\varphi & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \end{pmatrix} \qquad |A| = 1$$

$$\mathrm{adj}(A) = \begin{pmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \cos\theta\cos\varphi & \sin\varphi\cos\theta & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \end{pmatrix}^*$$

$$= \begin{pmatrix} \sin\theta\cos\varphi & \cos\theta\cos\varphi & -\sin\varphi \\ \sin\theta\sin\varphi & \sin\varphi\cos\theta & \cos\varphi \\ \cos\theta & -\sin\theta & 0 \end{pmatrix} = A^{-1}$$

$$\left[\vec{e}_R\cdot\dddot{\vec{D}}(t)\right]\times\vec{e}_R$$

$$= q\omega^3 a^2\sin\theta\Big[-\cos\theta\cos(2\omega t-\varphi)\,(\sin\theta\cos\varphi\,\vec{e}_R + \cos\theta\cos\varphi\,\vec{e}_\theta - \sin\varphi\,\vec{e}_\varphi)$$

$$-\cos\theta\sin(2\omega t-\varphi)\,(\sin\theta\sin\varphi\,\vec{e}_R + \sin\varphi\cos\theta\,\vec{e}_\theta + \cos\varphi\,\vec{e}_\varphi)$$

$$+\sin\theta\cos(2\omega t-2\varphi)\,(\cos\theta\,\vec{e}_R - \sin\theta\,\vec{e}_\theta)\Big]$$

$$= q\omega^3 a^2\sin\theta\Big\{\sin\theta\cos\theta\left[-\cos(2\omega t-2\varphi) + \cos(2\omega t-2\varphi)\right]\vec{e}_R$$

$$-\left[\cos^2\theta\cos(2\omega t-2\varphi) + \sin^2\theta\cos(2\omega t-2\varphi)\right]\vec{e}_\theta$$

$$-\cos\theta\sin(2\omega t-2\varphi)\,\vec{e}_\varphi\Big\}$$

$$= -q\omega^3 a^2\sin\theta\left\{\cos(2\omega t-2\varphi)\,\vec{e}_\theta + \cos\theta\sin(2\omega t-2\varphi)\,\vec{e}_\varphi\right\}$$

$$\left\{\left[\vec{e}_R\cdot\dddot{\vec{D}}(t)\right]\times\vec{e}_R\right\}\times\vec{e}_R$$

$$= q\omega^3 a^2\sin\theta\left[-\cos\theta\sin(2\omega t-2\varphi)\,\vec{e}_\theta + \cos(2\omega t-2\varphi)\,\vec{e}_\varphi\right]$$

$$\vec{B} = -\frac{\mu_0}{4\pi c^2}\,\frac{e^{ik_0R}}{R}\,\vec{e}_R\times\left[\vec{e}_R\cdot\ddot{\vec{D}}(t)\right]$$

$$e^{ik_0R}$$

$$= -\frac{\mu_0}{4\pi c^2} q\omega a^2 \frac{\quad}{R} \sin\theta \{\cos(2\omega t - 2\varphi)\,\vec{e}_\theta + \cos\theta \sin(2\omega t - 2\varphi)\,\vec{e}_\varphi$$

$$\vec{E} = \frac{\mu_0}{4\pi c^2} \frac{e^{ik_0R}}{R} \vec{e}_R \times \{\vec{e}_R \times [\vec{e}_R \times \dddot{\vec{D}}(\tau)]\}$$

$$= \frac{\mu_0}{4\pi c^2} q\omega^3 a^2 \frac{e^{ik_0R}}{R} \sin\theta \left[-\cos\theta \sin(2\omega t - 2\varphi)\,\vec{e}_\theta + \cos(2\omega t - 2\varphi)\,\vec{e}_\varphi\right]$$

$\theta = 0$：$\vec{E} = \vec{B} = 0$，无辐射。

$\theta = 90^\circ$ 磁场：线偏振，电场：圆偏振。

§ 6.4 天线与天线阵辐射

1. 对称偶直天线

设天线电流分布为 $I(z,t) = I_0 \sin k_0\left(\frac{l}{2} - |z|\right) e^{-i\omega t}$ $\left(|z| \le \frac{l}{2}\right)$

$$\vec{A}(\vec{R},t) = \frac{\mu_0}{4\pi} \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{1}{r} I\left(z, t - \frac{r}{c}\right) dz\, \vec{e}_z$$

$$= \frac{\mu_0}{4\pi} \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{1}{r} I_0 \sin k_0\left(\frac{l}{2} - |z|\right) e^{-i\omega\left(t - \frac{r}{c}\right)} dz\, \vec{e}_z$$

远场近似：$r \approx \vec{R} - \vec{e}_z \cdot \vec{R}' = R - z\cos\theta$

$$A(\vec{R},t) \approx \frac{\mu_0 I_0}{4\pi R} e^{i(k_0R - \omega t)} \int_{-\frac{l}{2}}^{\frac{l}{2}} \sin k_0\left(\frac{l}{2} - |z|\right) e^{-ik_0 z\cos\theta} dz$$

$$= \frac{\mu_0 I_0}{4\pi R} e^{i(k_0R - \omega t)} \int_{-\frac{l}{2}}^{\frac{l}{2}} \sin k_0\left(\frac{l}{2} - |z|\right) \left[\cos(k_0 z\cos\theta) - i\sin(k_0 z\cos\theta)\right] dz$$

$$= \frac{\mu_0 I_0}{4\pi R} e^{i(k_0R - \omega t)} \int_{-\frac{l}{2}}^{\frac{l}{2}} \sin k_0\left(\frac{l}{2} - |z|\right) \left[\cos(k_0 z\cos\theta) - i\sin(k_0 z\cos\theta)\right] dz$$

$$= \frac{\mu_0 I_0}{4\pi R} e^{i(k_0R - \omega t)} \int_{-\frac{l}{2}}^{\frac{l}{2}} \sin k_0\left(\frac{l}{2} - |z|\right) \cos(k_0 z\cos\theta)\, dz$$

$$= \frac{\mu_0 I_0}{4\pi R} e^{i(k_0R - \omega t)} M$$

$$M = \int_{-\frac{l}{2}}^{\frac{l}{2}} \sin k_0\left(\frac{l}{2} - |z|\right) \cos(k_0 z\cos\theta)\, dz \qquad \cancel{d(\sin z')} = \cos z'\, k_0(-dz)$$

$$= \frac{2}{k_0\cos\theta} \int_0^{\frac{l}{2}} \sin z'\, d\left[\sin(k_0 z\cos\theta)\right], \quad z' = k_0\left(\frac{l}{2} - z\right)$$

$$= \frac{2}{k_0\cos\theta} \left[\sin z' \sin(k_0 z\cos\theta)\Big|_{z=0}^{z=\frac{l}{2}} + k_0 \int_0^{\frac{l}{2}} \cos z' \sin(k_0 z\cos\theta)\, dz\right]$$

$$= \frac{-2}{(k_0\cos\theta)^2} k_0 \int_0^{\frac{l}{2}} \cos z'\, d\left[\cos(k_0 z\cos\theta)\right]$$

$$= \frac{-2}{k_0\cos^2\theta} \left[\cos z' \cos(k_0 z\cos\theta)\Big|_{z=0}^{z=\frac{l}{2}} - k_0 \int_0^{\frac{l}{2}} \sin z' \cos(k_0 z\cos\theta)\, dz\right]$$

$$= -\frac{2}{k_0\cos^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right] + \frac{1}{\cos^2\theta}M$$

解得 $M = \frac{\cos^2\theta}{\sin^2\theta}\frac{2}{k_0\cos^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]$

$$= \frac{2}{k_0\sin^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]$$

$$A(\vec{R},t) = \frac{\mu_0 I_0}{4\pi R}e^{i(k_0R-\omega t)}\frac{2}{k_0\sin^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]$$

$$= \frac{\mu_0 I_0}{2\pi k_0 R\sin^2\theta}e^{i(k_0R-\omega t)}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]$$

$$\vec{B}(\vec{R},t) = \nabla\times\vec{A}(\vec{R},t) = ik_0\vec{e}_R\times A(\vec{R},t)\vec{e}_z$$

$$= i\frac{\mu_0 I_0}{2\pi R\sin\theta}e^{i(k_0R-\omega t)}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]\vec{e}_\varphi$$

$$\vec{E}(\vec{R},t) = c\vec{B}(\vec{R},t)\times\vec{e}_R$$

$$= -i\frac{c\mu_0 I_0}{2\pi R\sin\theta}e^{i(k_0R-\omega t)}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]\vec{e}_\theta$$

$$\vec{S} = \mathrm{Re}\left[\vec{E}(\vec{R},t)\right]\times\frac{1}{\mu_0}\mathrm{Re}\left[\vec{B}(\vec{R},t)\right]$$

$$= \frac{c\mu_0 I_0^2}{4\pi^2R^2\sin^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]^2\sin^2(k_0R-\omega t)\,\vec{e}_R$$

$$\langle\vec{S}\rangle = \frac{c\mu_0 I_0^2}{8\pi^2R^2\sin^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]^2\vec{e}_R$$

$$= \frac{c\mu_0 I_0^2}{8\pi^2R^2}f(\theta)\,\vec{e}_R$$

$$f(\theta) = \frac{1}{\sin^2\theta}\left[\cos\left(\frac{1}{2}k_0 L\cos\theta\right) - \cos\left(\frac{1}{2}k_0 L\right)\right]^2$$

2. 半波天线 对称天线的长度为波长的半整数倍.

$L_m = \frac{m\lambda_0}{2}$, $k_0 L_m = m\pi$

此时 $I(z,t) = I_0\sin\left(\frac{1}{2}m\pi - k_0|z|\right)e^{-i\omega t}$

$$f(\theta) = \frac{1}{\sin^2\theta}\cos^2\left[\frac{m\pi}{2}\cos\theta\right]$$

例: 计算 $m=1$ 的半波天线的总辐射功率.

$I_1(z,t) = I_0\cos(k_0|z|)e^{-i\omega t}$

$$f(\theta) = \frac{1}{\sin^2\theta}\cos^2\left(\frac{\pi}{2}\cos\theta\right)$$

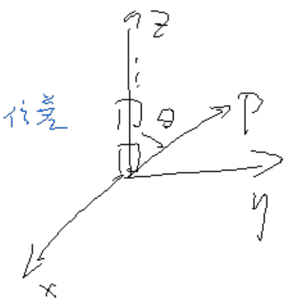
$$P = \oint \langle S\rangle R^2\, d\Omega$$

$$= \frac{c\mu_0 I_0^2}{8\pi^2 R^2} R^2 \int_0^{\pi} \frac{1}{\sin^2\theta}\cos^2\left(\frac{\pi}{2}\cos\theta\right) 2\pi \sin\theta\, d\theta$$

$$= \frac{c\mu_0 I_0^2}{4\pi}\int_0^{\pi}\frac{1}{\sin\theta}\cos^2\left(\frac{\pi}{2}\cos\theta\right) d\theta$$

$$\approx 1.22\,\frac{c\mu_0 I_0^2}{4\pi}$$

3. 半波天线直线阵 — 纵向.



相邻两天线的相位差

$\alpha(\theta) = k_0 d\cos\theta + \xi$ 天线间的等相位差

$$E = \sum_{m=1}^{N} E_0\, e^{i(m-1)\alpha}$$

$$= E_0\,\frac{1-e^{iN\alpha}}{1-e^{i\alpha}}$$

$$I(\theta) = \frac{1}{\sin^2\theta}\cos^2\left(\frac{\pi}{2}\cos\theta\right)\left|\frac{1-e^{iN\alpha}}{1-e^{i\alpha}}\right|^2$$

$$= f(\theta)\,\frac{\sin^2\left(\frac{N\alpha}{2}\right)}{\sin^2\left(\frac{\alpha}{2}\right)} = f(\theta)\,F(\alpha)$$

$\xi = 0$ 时：

(1) $\theta = \frac{\pi}{2}$, $F(\alpha) = F_{max} = N^2$

(2) $\theta \neq \frac{\pi}{2}$, 极小角 $\frac{N\alpha_m}{2} = m\pi$, $k_0 d\cos\theta_m = \frac{2m\pi}{N}$

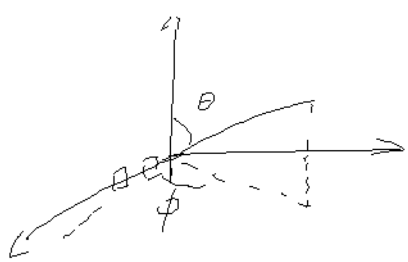
定义张角 $\psi = \theta_0 - \theta_1 = \frac{\pi}{2} - \theta_1$

$$= \frac{\pi}{2} - \arccos\frac{2\pi}{k_0 d N}$$

$$\sin\psi = \frac{\lambda_0}{Nd}$$

$N \nearrow$ $\psi \searrow$ 辐射集中

4. $\alpha(\theta,\phi) = k_0 d\sin\theta\cos\phi$



$$F(\alpha) = \frac{\sin^2\left(\frac{1}{2}N k_0 d\sin\theta\cos\phi\right)}{\sin^2\left(\frac{1}{2}k_0 d\sin\theta\cos\phi\right)}$$

主极大 $\sin\theta\cos\phi = 0$ $\quad \phi = \pm\frac{\pi}{2}$.